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Policy Brief: AI Safety Through Coordination Physics

For AI Alignment Researchers and Safety Organizations

Key Message

AI alignment is fundamentally a coordination problem. The goal of AI safety can be reframed as: maintaining civilizational trust (H_3) above the coordination threshold ($\theta \approx 0.375$) throughout the AI transition.

The Coordination Framing of AI Risk

Traditional AI Safety Concerns

- Misaligned objectives
- Power-seeking behavior
- Deceptive alignment
- Catastrophic misuse

- Race dynamics

Coordination Physics Perspective

All of these are **mechanisms by which AI could push H_3 below θ** :

AI Risk	Coordination Mechanism	Impact on H_3
Misaligned AI	Erodes trust in institutions	-0.05 to -0.15
Deepfakes	Destroys shared reality	-0.03 to -0.08
Manipulation	Fragments coordination	-0.08 to -0.15
Economic disruption	Breaks social contracts	-0.03 to -0.05
Information overload	Overwhelms verification	-0.02 to -0.05

The common thread: AI accelerates λ (modernization pressure) faster than it builds H_3 (trust).

Law 17: The AI Coordination Paradox

Core Insight:

AI amplifies coordination dynamics in BOTH directions.

$$dH_3/dt_{AI} = dH_3/dt_{natural} \times (1 + A_{positive})$$

$$d\lambda/dt_{AI} = d\lambda/dt_{natural} \times (1 + A_{negative})$$

Currently: $A_{negative} \approx 0.7 > A_{positive} \approx 0.3$

Result: AI is accelerating us toward the threshold.

Why This Matters for Alignment

Traditional alignment asks: "How do we make AI do what we want?"

Coordination physics asks: "How do we ensure AI maintains our capacity to coordinate?"

Key insight: An AI that is individually aligned but coordination-negative is still catastrophically dangerous.

Three Scenarios for AI-Civilization Interaction

Scenario	Probability	Path	Outcome
A: Collapse	40%	AI accelerates to θ	Civilizational fragmentation
B: Enhancement	25%	AI governance succeeds	Civilizational transcendence
C: Unstable	35%	Oscillates near θ	Repeated crises, eventual resolution

The goal of AI safety: Shift probability from A toward B.

Implications for AI Safety Research

1. Coordination Impact as Alignment Metric

Current practice: Align AI to human preferences, values, or instructions.

Recommended addition: Measure AI's impact on H_3 .

Coordination-Aligned AI = AI where:

$dH_3/dt_{\text{deployment}} > 0$

OR

$dH_3/dt_{\text{deployment}} > d\lambda/dt_{\text{deployment}} \times -0.5$

An AI that builds trust faster than it accelerates change is coordination-safe.

2. RLHF Extensions

Current: Reinforcement Learning from Human Feedback

Proposed: RLHF + Coordination Feedback (RLHF-C)

Additional training signal: - Does output increase cross-group understanding? - Does output maintain shared reality? - Does output support collective decision-making? - Does output avoid polarization amplification?

3. Constitutional AI Extensions

Add to constitutional principles:

COORDINATION PRINCIPLES:

1. I will not generate content that fragments shared reality
2. I will present multiple perspectives on contested issues
3. I will not amplify polarization

4. I will support cross-group understanding
5. I will enhance rather than replace human coordination capacity

4. Capability Evaluation Extensions

Add coordination impact to capability evaluations:

Capability	Traditional Metric	Coordination Metric
Language	Fluency, accuracy	Perspective balance, bridging
Persuasion	Effectiveness	Polarization effect
Assistance	Task completion	Trust maintenance
Information	Accuracy	Reality anchoring

Recommendations for AI Labs

Immediate (Next 6 Months)

1. **Add H₃ monitoring to deployment metrics**
 - Track polarization in AI-assisted conversations
 - Measure cross-group exposure
 - Monitor trust changes in user populations
2. **Incorporate coordination principles in training**
 - Add bridging behavior to RLHF
 - Penalize polarization amplification
 - Reward perspective-taking
3. **Develop coordination safety benchmarks**
 - Test AI for cross-group understanding support
 - Measure reality anchoring
 - Evaluate collective decision-making enhancement

Short-Term (6-18 Months)

1. **Coordination Impact Assessment before deployment**
 - Formal assessment of expected ΔH_3 and $\Delta \lambda$
 - Mitigation requirements for negative scores
 - Monitoring commitments
2. **Design for H₃ maintenance**
 - Fact-checking integration
 - Source attribution
 - Multiple perspective presentation
 - Uncertainty acknowledgment
3. **Governance pacing**
 - Match deployment speed to governance capacity
 - Incremental rollout with monitoring
 - Automatic throttling on coordination damage

Medium-Term (18-36 Months)

1. Industry coordination on safety standards

- Shared H_3 measurement protocols
- Common safety benchmarks
- Cross-company early warning

2. Regulatory engagement

- Support coordination-focused AI governance
 - Participate in international frameworks
 - Advocate for evidence-based policy
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The Alignment Reframe

From Individual to Collective Alignment

Traditional: Align AI to individual human preferences

Problem: Individuals may have coordination-negative preferences

Solution: Align AI to collective coordination maintenance

Individual alignment + Coordination alignment = Safe AI

From Capability Control to Coordination Maintenance

Traditional: Control AI capabilities to prevent misuse

Problem: Capability control is increasingly difficult

Solution: Ensure AI capabilities build coordination capacity

If AI maintains $H_3 > \theta$, many capability risks become manageable.

From Racing to Coordinating

Traditional: Race to develop AI faster than competitors

Problem: Racing creates pressure to cut safety corners

Solution: Coordinate on safety, compete on capability within safe bounds

Industry coordination on safety = Higher H_3 = Better outcomes for all

Key Insight

AI safety is not just about preventing AI from doing bad things. It's about ensuring AI preserves humanity's capacity to coordinate.

A world with perfectly aligned individual AIs that collectively fragment human coordination is not a safe world.

The goal: AI that maintains $H_3 > \theta$ throughout the transition.

Resources

- Full framework: Paper 2C, The 17 Laws of Civilizational Coordination
- Application to Earth: Earth Coordination Status Report 2025
- Technology guidance: Coordination Technology Framework
- Law 17 details: LAW_17_AI_COORDINATION_PARADOX.md

“AI alignment is ultimately about coordination. An AI that cannot maintain our ability to work together is not truly aligned, no matter how well it follows instructions.”